Master Software Engineering
Informatics Institute
Faculty of Science
University of Amsterdam

Master Project Software Engineering

Student Manual

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Contents

1. Introduction

The Master Project Software Engineering is a mandatory part of the master programme, in which you design and conduct a practical research project at the university or an external organisation. In the Master Software Engineering, the terms *dissertation*, *thesis*, and *scriptie* (Dutch) are used synonymously.

This manual gives more details on the objectives, requirements, and steps to take to a successful completion. For any questions regarding the master project, please contact the thesis coordinator.

This manual is organised as follows. We compiled the crucial information in chapter 2 and chapter 3 gives you a clear overview of the objectives and format. In chapter 4 you can find a list of common topics. Your project will have two artefacts: the master thesis and the defense. A detailed description of these can be found in chapter 5. To make the most of your time with your supervisor, chapter 6 provides a detailed meeting protocol. As the master project is one likely of the most challenging you will face, chapter 7 shows the various practical and mental support resources available, because you do not have to go at it alone. Finally, we have broken down the path to success in eight easy steps in chapter 8.

2. Important Information

In this chapter we provide a birds-eye view of the most important information regarding the master project. The master project itself is explained in more details in the next chapters.

2.1. Prerequisites

To be able to start your master project, you should have passed at least five courses, of which one is Preparation Master Project (PMP)[1]. To pass PMP, your project proposal needs to be approved by your academic supervisor and the thesis coordinator.

2.2. Master Project Timelines

The master project takes place according to the following recommended timeline. You have some flexibility on how you want to run it; it is your show, but be aware that all the course and support activities are scheduled to fit this timeline.

Timeline full-time

- Preparation Phase:
  - Project Acquisition: September – December (Block 1–2).
  - Proposal writing: January (Block 3).
  - Pitch sessions: last week of Block 3.
  - Proposal submission: end of Block 3.
- Execution Phase:
  - Thesis Kick-off session: April (first week of Block 5).
  - Project execution and thesis writing: April – June (Block 5–6).
  - Midterm presentations: May (end of Block 5).
  - Defense: June – August (end of Block 6 till end of academic year).

Timeline part-time For a part-time student, the timeline is roughly the same, expect you have a bit more flexibility. The general advise is to spread the workload and do the Preparation Phase in year 1, and the Execution Phase in year 2. Alternatively, you can choose to do both phases in year 2, as it may better align with the expectations of the host organisation, though at an increased workload. Depending on your circumstances,

[1] See Teaching and Exam Regulations, Part B, article B-4.6 – Sequence of examinations.

you may want to do the Execution Phase full-time in 3 months, instead of spreading the work over 6 months.

2.3. Writing Your Thesis

The master thesis should be written in English, and follow the LaTeX template as provided on Overleaf[2]. There is no page limit, though the thesis should be succinct and to the point. Brevity is appreciated!

2.4. Contacting the Ethical Committee

Responsibility for scientific integrity always lies first and foremost with the researchers themselves. You are expected to be attentive to ethical issues that may arise from your work. You are always required to reflect upon this starting in the early stages of your thesis process and discuss this with your academic supervisor. There should be an ethical reflection in your proposal, and in the discussion of your thesis.

Thereby, it is mandatory to fill in the ethics-self check via the RMS portal for the Faculty of Science[3]. If any issues arise, advice can be requested from the Ethics Committee, specifically ECIS-SP, the committee for student projects in Information Sciences. More information on the procedures and composition of the Ethics Committees can be found on the UvA staff page[4]. A request with the Ethics Committee may take up to eight weeks, so do not wait until the last moment to contact them.

2.5. Fraud & Plagiarism

The thesis you submit is checked for plagiarism using Turnitin Similarity. Your text is compared with those of fellow students and with published texts worldwide. This results in a percentage that indicates the extent to which your submitted assignment corresponds with sources on the internet, journals, and (previously) submitted documents by UvA students. The academic supervisor and second reader receive an overview of the results and thereby get an impression of the originality of your work.

In case of suspected plagiarism or fraud, you will be informed, and the Examinations Board will be notified. The Examinations Board will then investigate the claim and will give you an opportunity to present your side. The Examinations Board will then determine whether fraud or plagiarism occurred and impose appropriate sanctions, which can range from a written warning to permanently terminating your enrolment. See the "Regulations Governing Fraud and Plagiarism for UvA students"[5] to understand what the university exactly considers 'plagiarism' and 'fraud', and the possible sanctions.

[2] https://www.overleaf.com/read/zgjhspjzfjyw#615a77
[3] https://rms.uva.nl/servicedesk/customer/portal/14
[4] https://medewerker.uva.nl/en/science/content-secured/az/ethics-committees-faculty-of-science/faculty-of-science-ethics-committee-fec.html
[5] https://www.uva.nl/en/about-the-uva/policy-and-regulations/education/regulations-governing-fraud-and-plagiarism.html

2.6. The Use of (Generative) AI Tools

The UvA's fraud and plagiarism regulations emphasise the importance of lecturers being able to assess students' knowledge, insight, and skills. Consequently, we expect you to write everything you submit yourself. Unless explicitly stated otherwise, the use of (Generative) AI tools, such as ChatGPT, is therefore not permitted (also see Policy framework and guidelines on GenAI in education[6]). Submitting assignments that you did not write yourself may be considered plagiarism, and the UvA will take strict action in such cases.

While AI tools can be beneficial study aids, it is essential to be aware of the potential risks involved. For instance, ChatGPT can be used for brainstorming, checking your knowledge, or translating text. It can serve as a convenient and accessible assistant. However, it is crucial to understand that ChatGPT generates output based on probabilities and statistics, providing plausible answers rather than verifying the accuracy of the information. The output may be influenced by harmful biases and stereotypes, as the data sets used by ChatGPT are not always representative.

Furthermore, many AI tools store user interactions, posing privacy and intellectual property concerns. Therefore, it is advisable to avoid entering any privacy-sensitive or confidential information, such as confidential research data, patient information, or personal data of fellow students or lecturers. Even a paid account may not provide adequate security and privacy safeguards.

[6] https://www.uva.nl/en/about-the-uva/policy-and-regulations/education/policy-framework-and-guidelines-on-genai-in-education.html

3. Objectives

The objectives of the master project are explained in this chapter.

3.1. Focus

The focus of the master project is the scientific study of a problem relevant to academic and societal issues within the domain of software engineering. The project offers you the chance to collaborate with esteemed scientists or industry experts for an extended period, providing hands-on experience in research. The primary objective is to equip you with practical experience in applying both quantitative and qualitative scientific research methods and fostering your ability to work independently. Furthermore, the project serves as an opportunity for you to identify knowledge or skill gaps and address them.

Upon completing the master project, you are expected to possess the following learning outcomes:

- Formulate a clear research question in the field of software engineering and design a comprehensive plan to address it.
- Demonstrate a comprehensive understanding of the research project's area based on relevant literature, effectively applying this knowledge in practical scenarios.
- Analyse and critically evaluate research data, ensuring that the results align with the project's goals and research line.
- Communicate the research process and findings in a well-written report, including a detailed methodology and substantiated original phrasing.
- Present and discuss the results to both scientific and non-scientific audiences.
- Demonstrate proficiency in functioning effectively in a professional environment.

These learning outcomes are assessed based on three aspects: your research process, the master thesis, and the defense.

Justification It is crucial to present your research as a product. The thesis should not contain the research process itself, *i.e.* it is not a chronological log of all the actions you took. You must justify your actions and design choices based on merit. Your goal is to argue that your approach is a productive way of answering your research question, while contextualizing it in existing research. Avoid making a mistake into an excuse. Instead, transform it into a strategy for future research!

3.2. Research Format

Due to the interdisciplinary nature of the Software Engineering programme, the types of master's theses vary. Easterbrook et al. [1] provides a catalogue of empirical methods suitable for the Software Engineering domain, and describes the goals and the types of research question each best addresses. The most common research formats are:

- Empirical research, as used in social and economic science.
- Analysis of existing theories regarding a software engineering issue.
- Design/prototype of a complex software engineering method, or a new design approach or new algorithm for such a method.
- Evaluation of a complex software engineering system.
- Report with policy recommendations regarding a complex organisational issue, where the recommendations are explicitly based on theories of the software engineering domain.
- Experimental evaluation and/or specialisation of a design approach or method for a specific application domain.

4. Topics

The topic of the thesis should fit in the field of software engineering, focussing on the designing, developing, testing, or maintaining of software applications, in the broadest sense. Any topic you pick *must* push state of the art/state of practice.

If the topic is outside the scope of software engineering, the thesis coordinator must approve the proposed project before the thesis trajectory begins, to make sure it meets the programme's expectations.

4.1. Common Topics

Given software engineering is applicable to many research fields, it sometimes can be hard to determine whether a particular topic is truly related to *software engineering*. As a guideline, topics covered by SWEBOK v4 [2] are relevant topics for a thesis. Common topics are:

- **Software Requirements:** expression of the needs and constraints on a software product, or the activities necessary to develop and maintain the requirements for a software product and for the project that constructs it [2, Ch 01]. Example topics are Behaviour-driven development (BDD), automated requirements elicitation, requirements traceability in agile environments, *etc.*
- **Software Architecture:** the art and science of constructing software-intensive systems [2, Ch 02]. It can be considered part of Software Design, though often its more about the organisational structure of a software system. Typically you find subjects here like Model-view-controller (MVC), Representational State Transfer (REST), *etc.*
- **Software Design:** the basic concepts, context, and processes in which software requirements are analysed to define the software's external characteristics and internal structure as the basis for the software's construction [2, Ch 03]. Typically you find subjects here like Application Programming Interface (API), Unified Modelling Language (UML), design patterns.
- **Software Construction:** the detailed creation and maintenance of software through coding, verification, unit testing, integration testing and debugging [2, Ch 04]. This is closely linked to Software Design and Software Testing. Example subjects include low-code/zero-code platforms, dependency management, refactoring, cross-platform development, Domain-Specific Languages (DSLs), *etc.*
- **Software Testing:** dynamic validation of a System under test (SUT) to ensure it provides expected behaviours on a finite set of test cases, which are suitably selected from the usually infinite execution domain [2, Ch 05]. Examples here are unit testing, model-based testing, mutation testing, test coverage, *etc.*

- **Software Engineering Operations:** the set of activities and tasks necessary to deploy, operate and support a software application or system, while preserving its integrity and stability [2, Ch 06]. DevOps, MLOps, Platform Engineering are examples of activities that falls under this topic.
- **Software Maintenance:** the modification of software after delivery to correct faults, improve performance, or adapt to changes [2, Ch 07]. Common subjects are Continuous Integration/Continuous Delivery (CI/CD), maintainability, (impact of) code smells, predictive models for software maintenance effort estimation, *etc.*
- **Software Configuration Management:** the control and management of changes to software products throughout their lifecycle [2, Ch 08]. Example subjects are automated conflict resolution, impact of configuration management on DevOps, blockchain-based approaces to software version control, *etc.*
- **Software Engineering Management:** covers the planning, coordination, measuring, monitoring, controlling and reporting in software engineering projects. Example topics are metrics for distributed teams, risk management frameworks, impact of remote work, *etc.*
- **Software Engineering Process:** covers the definition, implementation, assessment, measurement, and improvement of software processes [2, Ch 10]. Example topics are Agile, Rational Unified Process (RUP), DevOps, process mining *etc.*
- **Software Engineering Models and Methods:** the various models, methods, and notations used in software engineering activities [2, Ch 11]. Example subjects are formal verification, model checking, UML, Systems Modelling Language (SysML), *etc.*
- **Software Quality:** explores concepts, techniques, and practices for ensuring and evaluation software quality [2, Ch 12]. Examples are automated quality assessment, impact of technical debt on software quality.
- **Software Security:** in addition to correctness and reliability, security of software has become an important quality [2, Ch 13]. Typical subjects are agile practices for software security, vulnerability checking tools, security patterns, *etc.*

4.2. Artificial Intelligence

Artificial Intelligence (AI) is an exciting and rapidly evolving field, and its influence is spreading across many research fields, including software engineering. It is crucial, however, to properly align your topic with the core research area of software engineering. Your thesis should focus on how AI relates to the creating, management and maintenance of software systems, not on developing new AI models from scratch. For example, a project on inventing a new deep learning algorithm, or improving an existing Large Language Model (LLM) is more suitable for a master's programme in AI.

The application of AI within software engineering broadly falls into two categories: 'AI for SE' and 'SE for AI'.

4.2.1. AI for SE: AI as a Tool

AI for SE is the application of existing AI models and techniques to solve problems within software engineering. The focus is on using AI as a tool to improve the software development process.

Examples of AI for SE projects include:

- Analyse developer productivity when using an AI-powered coding assistant.
- Applying natural language processing to automatically extract requirements from user feedback.
- Using machine learning to predict and prevent software defects.

For these projects, you are the software engineer, and *AI is a tool* you use. You are not expected to create or train new AI models. You are expected to demonstrate how existing ones can be integrated and evaluated to solve a specific software engineering problem.

4.2.2. SE for AI: AI as an Application Domain

SE for AI is the inverse, and refers to applying the methods and techniques developed within software engineering in the development and exploitation of AI systems. The goal is to make these systems robust, scalable, and maintainable.

Examples of SE for AI projects include:

- Designing a scalable architecture for a machine learning pipeline that can handle large datasets.
- Developing and testing a CI/CD framework for an AI-enabled application.
- Applying software quality assurance techniques to evaluate and ensure the reliability of an AI model in a production environment.

For these projects, you are ensuring that *the AI system is engineered properly.* While you will not be developing new models, you will need to understand and analyse their behaviour and training results to properly evaluate the software system you are building around them.

5. Artefacts

Within your master project, you will produce two artefacts you will be assessed on: the master thesis and the defense presentation. This chapter provides more details about both.

5.1. Master Thesis

A major artifact of your research project is the master thesis. A master thesis is defined as an individually written record of the your original research or design of a scientific nature. It is a unique and original piece of work, specially crafted for this occasion, reflecting your creative ideas. The thesis must support claims, hypotheses, policy recommendations, and design choices with arguments based on existing theory or empirical evidence.

The master thesis cannot be composed of copied resources (internet, books, journals, *etc.*) unless they are properly cited. The material should not have already been submitted elsewhere (in other courses, study programmes, or universities) with the intention of receiving study credits for this thesis. However, the thesis can expand upon previously submitted work, as long as it is clearly indicated which student contribution corresponds to which study program component.

There are a few requirements for the thesis, and a few expectations regarding organisation and contents. Depending on your research and approach, you may want to deviate from these guidelines, but it is always a good idea to discuss this with your academic supervisor.

5.1.1. Requirements

The thesis should be written by you in English (and not partially or fully generated, see section 2.6). British English is preferable, but American English is also acceptable, just make sure you are consistent. The thesis should use the LaTeX template as provided on Overleaf[1]. There is no word limit or target, though the thesis should be succinct and to the point. Brevity is appreciated!

5.1.2. Outline

A thesis, and for that matter any scientific communication, has a particular structure which we expect you to follow. This structure helps you in communicating your results, and it fits the readers' expectation, so they are not surprised (your thesis should not

[1] https://www.overleaf.com/read/zgjhspjzfjyw#615a77

read like a murder-mystery). Of course, these are guidelines, and depending on your research, you can deviate from this structure. Do discuss this with your academic supervisor first!

You can find a detailed description of the outline in the template, but the main elements are as follows:

Abstract
: This succinctly describes the context, problem and proposed solutions and conclusions. It should be at least 200 words, but no more than a page and avoid any citations.

Introduction
: The Introduction chapter itself is to provide your reader with the context of your research, and the problem you are trying to solve. It typically contains the following sections:

  Context
  : Provides the context of your research, and explains the importance and relevance of your topic within the field of software engineering.

  Problem Statement
  : Gives a concise description of the challenge you aim to address. The goal is to define the problem in clear terms, establish its significance, and provides a foundation for developing objectives, methods, and potential solutions. Be specific and avoid vague language.

  Research Questions
  : Explicitly lists the research questions or hypotheses you address in your research. It is helpful to enumerate them, for example as *RQ1*, *RQ2*, so you can refer to them in later chapters.

  Approach
  : Gives a brief overview of the approach (research methods) you will take to answer the research questions. This will provide the reader with a high-level roadmap.

  Outline
  : Gives the reader a short overview of the structure of your thesis, and what to find in each chapter.

Background
: The Background chapter provides information that has benefit for an informed audience. Often you use this to explain a technique or a concept you use, referring to literature. This is different from Related Work, where you show the relation of literature to your contributions. As a rule of thumb what to include, your thesis should still make sense without the Background chapter.

Related Work
: The Related Work chapter describes how other people have dealt with similar problems. The role of related work is threefold: grounding in earlier work, comparing your results with others, and inform the reader on state-of-the-art, and that you know what is ‘for sale’. Depending on your story, you may want to put

this chapter at the end of your thesis (before Conclusions) or at the beginning (after the Background).

Methods

Describes the design of your research, which methods and procedures you used. It should highlight the plausibility of your research methods, and clarify *why* you have chosen specific methods and how they are justified for your research.

Experiments & Results

Depending on the nature of your research, this is where you explain how you setup your experiments, and the results you got. In the case of an implementation (framework, prototype), you can describe the architecture and tools used, and the testing process used to show the validity and performance. This chapter is crucial for reproducibility.

Analysis

This is the most important part of your research: analysing the results you got in the previous chapters. Here you discuss the implications of your results, and how they answer the research questions. Be sure to cover *validation* in your analysis: how do you know the results you got are true.

Conclusion

This typically the last chapter you write, but the first chapter most people will read (after the abstract). Make sure to be concise and complete, but not too detailed. The interested reader can find that in the other chapters. The Conclusion chapter typically consist of the following sections:

Summary & Findings

Summarise your work (context, questions, approach) and present the findings in context with your research questions.

Contributions

Make explicit any contributions your thesis provides. These are tangible results, which can live outside the context of the thesis.

Limitations & Threats to Validity

No work is perfect. Discuss the limitations of your study, as a research study can only be as unbiased as the research and the circumstances you are working with. The threats to validity offers insights into the variables you encountered, and how you compensated for them.

Future Work

This is where you highlight all the work you wanted to do, but did not have the time. This is also where you can give an idea how to address any limitations you mentioned in the previous section.

Appendices

The appendices contain any non-crucial information. Often you put here tables and figures that are too big and do not contribute to the results and evaluation. Same with longer code examples and design files.

5.2. The Defense

The defense is the culmination of all your hard work. It is the moment where you present your research, discuss it with the committee, and receive the final grade for your project. Reaching this stage typically indicates that you have met all the necessary requirements for a passing grade.

The committee for your defense consists of your academic supervisor as chair, and the second reader. Any other supervisors (external, daily) can also be part of the committee, but they have an advisory role in the grading, in particular the research part of the rubric.

Your defense is public, and you can invite friends and family to join. The exam committee may also decide to join, to check whether everything goes according to the regulations.

The structure of the defense partially depends on your academic supervisor though in general it follows the format outlined below.

5.2.1. Presentation

You present your work to the committee and the audience in a 20–30 min presentation. Typically you will cover the context of your research, and the questions you started out with. After this you present your approach, the main results and their implications. You can conclude with a summary, and any future work. Depending on the nature of your work, you may use a different structure, or demonstrate the tool or framework you developed.

The level of your presentation should be around that of a master software engineering student. When introducing your research problem, however, it is helpful to present it in a way that the general audience (friends and family) understands why you spent all those months working hard. Although we do not expect them to understand all the details, it is important to be able to communicate the significance of your research interests to interested laypersons.

Do not make your presentation an exact copy of your thesis. This is a different medium, and often the structure and contents of a thesis do not translate well when presented as-is. Think about the audience and what the message is you want them to take away from it, *e.g.* what are the most important insights. When presenting graphs, be sure to explain the axis and highlight any important trends or key points. Often it is better to recreate the graphs you used in your thesis for the presentation (size, contrast) for visibility and clarity.

5.2.2. Q&A

Once the presentation is over, the questions and answers section of the defense starts. The chair of the committee can invite the audience to provide any questions. After which the committee takes over. The order is usually whoever travelled the furthest, which typically is the external supervisor, but this is all up to the chair.

The committee can ask questions to their heart's content. There is no time limit, though in practice it is typically around 30 minutes. The goal of the questioning is to assess your understanding of the project, and the knowledge you have gained. You can expect questions seeking clarification on your presented work, as well as hypothetical questions about the implications of your results in other contexts. Do not be afraid to ask clarifying questions, or start explaining your thought process. This part is meant as a discussion, or an exchange of ideas, not a cross-examination.

5.2.3. Deliberation

Once the committee is satisfied, they will retreat for deliberation, either by ushering everyone out of the room, or moving to a different location. During this deliberation, the committee will discuss their impression and assessment of the work. They will fill out the assessment form on DataNose according to the rubric as shown in appendix A.

5.2.4. Grade and Motivation

Once the committee has established the grade, they will call you in or return to the room, to tell you the final verdict and their motivation. You can choose to have this happen in public or private. The academic supervisor will typically give a description of the three categories, their assessment, the grade you got for each component, and the final grade. Please not that processing the grade may take a day or so to show up in SIS, after which you request your diploma (if this was your last course).

6. You and Your Academic Supervisor

Successfully navigation the relationship with your academic supervisor is a critical component of completing your master project. This chapter provides a protocol for your meetings and interactions, to ensure a productive and positive experience.

6.1. First Meeting

The first meeting is crucial for setting expectations and establishing the ground rules for your collaboration. Be prepared to discuss:

- Your desired grade: Do you simply want to pass? Are you aiming for a cum laude? Do you want to continue in academia, *e.g.* as a PhD student? What grade you aim for determines the required quality of your thesis, and whether you should be expected to write a (workshop) paper.
- Mutual expectations: What do you expect from your supervisor in terms of guidance, feedback, and availability? What does your supervisor expect from you in terms of independence, progress, and communication?
- Communication plan: Establish the frequency and format of your meetings, *e.g.* weekly, bi-weekly; in-person or virtual. Be sure to set up a standing calendar event, and do not cancel even if there is nothing to report. Agree on the best way to communication between meetings (*e.g.* e-mail, Slack, Teams).
- Project scope and goals: Come with a preliminary understanding of your research topic. Discuss the high-level goals, potential research questions, and the expected scope of the thesis.
- Roles and responsibilities: Clarify your role in driving the project and your supervisor's role as a guide. You are the project lead, your supervisor is your mentor and consultant.
- Holidays: Since the end of the master project is near the holiday period, it is good to know your supervisor's holiday plans, so you can consider that in your planning.

6.2. Regular Meetings

To make the most of your limited meeting time, preparation is key. How exactly you should prepare depends on the preference and agreement with your supervisor, but typically it is a good idea to:

- Send an agenda: A day or two before the meeting, send your supervisor a brief agenda. This should include:

  - A (brief!) summary of your progress since the last meeting.
  - A list of specific questions or problems you are facing.
  - A plan for what you intend to work on next.
- Share your work: Send any relevant materials well in advance. This could be a draft of a chapter, experimental results, a short presentation. This gives your supervisor time to review your progress and provide feedback.
- Review previous notes: Re-read the notes and action items from your last meeting to ensure you addressed them.

6.3. During the Meeting

Your meeting is a valuable opportunity for discussion and feedback. Make sure you make the most of it.

- Start with summary: Begin by briefly summarising what you have accomplished.
- Focus on problems: Clearly articulate any challenges and roadblocks. This is where your supervisor's experience is the most valuable. Instead of just saying "I'm stuck", explain what you have tried, what the results were, and what your hypotheses are.
- Discuss, do not just report: Engage in a discussion. Be open to feedback and constructive criticism. Assume your supervisor has your best interest, and is not "out to get you". It is okay to (respectfully) disagree with your supervisor's suggestions, but be prepared to justify your reasoning with evidence or a clear (technical) argument.
- Take detailed notes: Write down key takeaways, suggestions, and *especially* action items you agree upon for the next period.
- Confirm next steps: Before the meeting ends, summarise the action items for both you and your supervisor. This ensures you are both on the same page about the plan moving forward.

6.4. Effective Communication

Clear and consistent communication is the foundation of a good relationship:

- Be honest and transparent: If you are falling behind or struggling, communicate this early and honestly. We have all been there, we all know that it is hard. Hiding problems will only makes it worse.
- Keep a record: After each meeting, send a brief follow-up email summarising the key discussion points and agreed-upon action items. This helps prevent misunderstandings.
- Take ownership: This is your project. While your supervisor provides guidance, you are responsible for the progress and final product. Show initiative by proposing solutions to problems and suggesting directions for your research.

7. Support

The master project is a challenging project, and probably the first time you work on something by yourself for an extended time. Because research is inherently uncertain, it can sometimes be hard to define 'success' and to stay motivated and productive. This in itself is an important skill to develop, and one of the aspects you will be assessed on.

You do not have to do this all by yourself. There is a wide variety of support options, ranging from practical help with writing or math skills to more personal support for motivation and overcoming procrastination.

7.1. Academic Supervisor

First and foremost, your academic supervisor is there to support and guide you through this process. However, it is good to realise that this is your project, and you are in charge. Therefore, be proactive in communication with your supervisor and signal any challenges early. Also see chapter 6 for advice on building a collaborative and productive relationship with your academic supervisor.

7.2. Teaching Activities

Besides the academic supervisor, the master software engineering organises various events to help and support. During January, there are workshops to help you with write your proposal. At the end of January, you will participate in the Pitch Presentations, which gives you a chance to present your proposal and crystallise your thoughts. At the end of block 5 (end of May), the midterm presentations provides an opportunity to reflect on your progress and plan your next steps. Weekly stand-up meetings on Thursdays in block 5 (April – May) and block 6 (June) offer a chance discuss your progress and and any difficulties with your peers.

7.3. Help with Writing, Methods, or Statistics

The UvA has various resources and support options if you need help with writing, methods, or statistics.

7.3.1. Writing

The Writing Centre[1] offers free courses, workshops and individual tutoring sessions for all UvA students. Of particular interest is 'Thesis Week' for students who are struggling to work on their thesis. For a week, you will meet twice a day with a writing tutor and a small group of other students to make a plan for the day, discuss your progress, and get advice and support. More information on how to sign up can be found on the page on Writing Centre Thesis Week[2].

7.3.2. Methods and Statistics

The Psychology Methodology Shop gives advice on questions and bottlenecks you may run into in your research. This shop is primarily intended for psychology students, but are also open to students from other programmes. They offer useful sources you can use to tackle issues regarding your research design and analysis. If you cannot resolve issues on your own, you can drop in during the walk-in clinics for support. See Help with methods and statistics[3] for more information, and how to book a time slot (as it tends to get busy).

When you experience problems with mathematics or statistics, you can get help from the Statistics and Mathematics Advice and Support Hub (SMASH). Friendly tutors offer both in-person and online drop-in sessions, where you can get help help with specific problem or clarification on mathematical topics. You can check Canvas[4] for resources and office hours, or contact them via `smash-science@uva.nl`.

7.4. When you run Into Problems With Your Thesis

When you are struggling with motivation or procrastination, the Guided Study Session[5] organised by the Library can be an excellent way to get you back on track. During the guided study sessions, you study in a group with support from student hosts and specialists. The hosts are trained by student counsellors and student psychologists. The specialists are experts from the Writing Centre, Methodology Shop, thesis coaches, or information specialists from the Library. The Library offers four types of sessions, including AD(H)D-friendly sessions, thesis sessions, and a summer study sprint.

If you are experiencing problems with your thesis for an extended time, consider joining the Graduation support group[6]. This weekly support group supports you by getting you in contact with fellow students in similar circumstances. Together, you agree on realistic targets. The group offers you structure and organisation. Furthermore, you can benefit from the group as a positive stimulus and an incentive to get things done. These sessions are offered both in-person and online.

[1] https://intt.uva.nl/language-training/writing-centre/writing-centre.html
[2] https://intt.uva.nl/language-training/writing-centre/thesis-week.html
[3] https://student.uva.nl/en/information/help-with-methods-and-statistics
[4] https://canvas.uva.nl/enroll/RLMWBN
[5] https://student.uva.nl/en/information/guided-study-sessions
[6] https://student.uva.nl/en/information/graduation-support-group

If you run into problems with the thesis itself, you can contact the thesis supervisor. For personal issues, arrange a meeting with the study advisor as soon as possible. Together with them, you can draw up a plan to successfully finish up your thesis. If you need to see a psychologist, contact the student psychologist[7] immediately. If you are unsure which help would be best for you, start by discussing this with study adviser.

[7] https://student.uva.nl/en/information/uva-psychologists

8. Eight Easy Steps to Success

Below, you will find general information on what we expect from you. The step-by-step guide will introduce you to the master thesis process. Read this information carefully! More details can be found in the previous chapters. If you have any questions, please contact the thesis or programme coordinator.

8.1. Step 0: Before you Start

You need to register for the course *Preparation Masterproject Software Engineering* (PMP) and *Masterproject Software Engineering*. The registration for this course follows the same procedure for other courses which you can find under *course registration* in the Student guide[1].

You may start with your master project (step 4) if you have completed at least five courses, of which one must be Preparation Master Project[2] (PMP). To pass PMP, your project proposal must have been approved by your academic supervisor and the thesis coordinator.

8.2. Step 1: Choose a Master Project

The first step in a successful master project, is finding a topic you are interested in. There are several ways and events where you can find a project, which are explained in the following sections. To keep your studies on track, you should have secured a thesis topic by the end of block 2, *i.e.* before Christmas.

8.2.1. Internal Projects

Many of the research groups that are part of the Informatics Institute have projects available for you to join and write your master thesis on. These projects can be found on the Thesis Matching Platform[3] and will occasionally be posted on the Canvas page. One advantage of choosing an internal project is that they come with an UvA academic supervisor, and the project is available and acceptable. Often these projects are more flexible to fit into your schedule.

[1] https://student.uva.nl/en/topics/course-registration
[2] See Teaching and Exam Regulations, Part B, article B-4.6. – Sequence of examinations
[3] https://datanose.uva.nl/m/thesismatching

8.2.2. External (Internship) Projects

Some students do a thesis as an internship. These projects can be found on the Thesis Matching Platform[3]. For projects that are not on the Matching Platform, it is strongly recommended to ask the thesis coordinator for permissions, to make sure it meets the Master's expectations, and to avoid any disappointment if your proposal gets rejected.

When you do an internship, it is required that the UvA academic supervisor and second reader have access to the code and collected data to guarantee the quality of assessment. If this yields a problem, the thesis project cannot continue.

Know that an external project (or any master project) should be about answering a research question with scientific relevance. It is not acceptable to simply deliver a practical solution for the organization or company involved. Also, it should not be about gaining work experience, as may be typical with an internship. Make sure you get these expectations clear on all sides.

All onboarding information for (external) supervisors can be found on Canvas via the course Preparation Master Project → Modules → Guides → Onboarding for Thesis Supervisors. If you are doing an external project, you will also have to find a UvA supervisor for supervision. In case you cannot find a UvA supervisor, an external project cannot be used as a thesis project.

8.2.3. Proposing a Project Yourself

It is possible to suggest a project of your own. In that case, you have to seek approval from the thesis coordinator, as the project needs to be at a Master's level and needs to conform to the set requirements. Approval on self-found company can only be given based on a project proposal following the template. The same applies if you want to do your thesis project at the company you work for. We do not, however, recommend that you do your project at your own workplace. Our experience is that it will lead to delay or a stranded project, since thesis work and normal work are hard to distinguish. If you want to persist and you get approval to go on, please make sure that you do the thesis at another department than where you work. If you have any questions, please contact the thesis or programme coordinator.

8.2.4. Thesis Fair – October

The Thesis Fair is held in October for all the master programmes of the Informatics Institute of the UvA. During the fair, you will meet organisations for an internal or external project. Organisations and university research groups will all offer projects that you can use for your thesis project. The organiser of the thesis fair (`thesisfair-IvI@uva.nl`) will contact you in September to invite you to an information session (please make sure you join) to explain how everything works. Invitations will be sent out via e-mail and Canvas. All projects will be posted on the Thesis Matching Platform. You can contact the research group directly if you find an interesting project. External organisations can be contacted at the Thesis Fair, or directly if their projects are marked as "Matching Platform only" (*i.e.* *without* double asterisks ('**') in their project title).

8.2.5. Thesis Borrel – November

The Thesis Borrel is a social event held in November, only for master Software Engineering students. At this event, researchers from the various research groups will present their research and project proposal. This is the ideal event to pick up a research project (besides the Thesis Fair), and meet your potential academic supervisor.

8.2.6. Contracts and/or NDAs

The programme sees that many external companies or organisations request students to sign a contract and/or a NDA (non-disclosure agreement) to protect their valuable data and IP. We understand the importance for this, but urge you to not sign any contract, as the terms may be unreasonable punitive (*i.e.* demanding a 1 million euro fine when breaking the terms), or prevent you from graduating altogether (*i.e.* by prohibiting the presentation of the results at your public defence).

To that end, Universiteiten van Nederland (UNL), in collaboration with industry, have set up a model internship contract that protects the company's IP, while still allowing you to graduate. The model contract can be found on the student information site on internships[4]. The UvA does not sign any contract other than the approved university (UNL) contract. You can use this contract and get it signed by the Programme Director. Be aware that it may take up to eight weeks before the contract is signed by the university, so plan accordingly. If you have any questions about this contract, or what to make modifications, please contact the internship agreement coordinator via `internshipagreement-science@uva.nl`.

8.3. Step 2: Find an Academic Supervisor

After choosing a master project, you have to find an academic supervisor who can support you. Internal projects often already come with an academic supervisor, so you are done with this step. For external or self-proposed projects you will have to find an academic supervisor. Canvas, PMP → Modules → Guides → Finding an Academic Supervisor[5] for an overview of the research groups and their area of expertise.

The academic supervisor has to be on the list of examiners, which you can find via DataNose → Programme List → Master Software Engineering → List of examiners[6]. If your prospective supervisor is not on the list, send the thesis coordinator a request for addition. The academic supervisor has to be a permanent staff member of the UvA with a PhD. They should also have an UTQ certificate, or are in the process of obtaining one. Exceptions can be made for well-motivated cases.

[4] https://student.uva.nl/en/topics/internship-possibilities-and-procedure
[5] https://canvas.uva.nl/courses/52891/files/13671007?module_item_id=2582212
[6] https://datanose.nl/#program[NMA_SE]/examiners

8.4. Step 3: Write your Thesis Proposal

January is all about writing your proposal. Think of it as the foundation for your thesis. It will give you a clear idea of what needs to be done and help you plan. It is also your chance to show your supervisor that your project is worth their time and resources. Plus, it sets clear expectations for both you and your supervisor.

Write your proposal using the provided Overleaf proposal template[7]. The proposal should a problem statement, research question(s), literature review, methods, risks, ethical considerations, and a timeline. To craft a strong problem statement, spend considerable time reviewing research related to your topic to get a clear understanding of the state-of-the-art. In your problem statement, explain what has already been done in this field, identify the existing issues, and describe how your research will address these problems. Check out the Preparing your literature study[8] section on Canvas for more guidance on how to find relevant literature.

Once you have identified your research gap, design research questions that will help address and fill this gap when answered. While there are different types of research questions[9], they all need to be answerable, based on previous research, clearly focused, and aimed at adding new knowledge. Remember, your research questions do not need to cover the entire research gap-focus on a specific aspect that aligns with your resources, time, and methods. You can refine your research questions as you go, but the core principles should stay the same.

In the research method section, explain how you will go about answering the question. Different types of research questions require different research methods[10].

Next, identify potential risks that could arise during your project. While you may not foresee all risks, try to anticipate as many as possible to prepare for potential challenges. For example, a risk might be that the gap between your current expertise and the expertise needed could lead to delays.

For the ethical consideration, you need to fill out the ethics self-check form on the RMS portal of the Faculty of Science, and present the summary of the results, and any further actions. If you want, you can consult the Ethical Committee, specifically ECIS-SP for SE students, for advice. Know that a request for advice can take up to eight weeks, so do not delay.

Finally, break your project into smaller, manageable parts and create a timeline. This is not a strict schedule, but a way to get a clearer view of how much time you have and how to use it effectively.

8.4.1. Tips

In the PMP and Master Project Proposal module on Canvas, you will find additional information including previously written proposals, research questions and methods, and more.

[7] https://www.overleaf.com/read/ppjrknfrcqxw#e90553
[8] https://canvas.uva.nl/courses/52891/pages/preparing-your-literature-study
[9] https://canvas.uva.nl/courses/52891/pages/research-methods
[10] https://canvas.uva.nl/courses/52891/pages/research-methods

To organise the literature you have read, consider using dedicated reference management software like Zotero[11], Mendeley[12], or BibDesk[13]. BibLaTeX[14] will also help you manage the bibliography in your proposal and thesis.

8.4.2. Pitch Presentation – January

At the end of January you are going to give a short (about 5 min) pitch presentation of your proposal. Prepare a few slides that explain the problem behind your topic, your (draft) research question(s), and your approach to answer these questions. Include all the necessary information, but remember to keep it concise since you only have 5 minutes. This presentation is an opportunity to get initial feedback on your project from a wider audience. It is also a chance to raise any questions or uncertainties you might have.

8.4.3. Approval

The research proposal, including a state-of-the-art survey for the corresponding research topic, has to be approved by the academic supervisor and the thesis coordinator. If these roles point to the same person, the role of the thesis coordinator is delegated to the programme director. If all these three roles point to the same person, we delegate the role of the programme director to an Examinations Committee member.

The official approval from the academic supervisor and thesis coordinator is given via DataNose, where you should create an entry for your thesis project. See the Canvas page on "Submitting your project proposal in DataNose"[15] for details. For questions on how to do that if not immediately clear from the DataNose interface please contact `datanose-fnwi@uva.nl`. Make sure your academic supervisor agrees before submitting, as you cannot change the proposal PDF after submission.

The deadline for submission is at the end of Block 3, after the pitch presentation sessions. When approval is received in DataNose, the submitted proposal is marked as passed and you receive the 6 ECTS for PMP. These ECTS only participate in the number of ECTS required for graduation, they do not impact your grade average.

Common Questions

I don't have an academic supervisor (examiner) yet You cannot submit your proposal if you don't have an academic supervisor yet. Email the thesis coordinator to let them know you are still looking for a supervisor.

My academic supervisor (examiner) is not on the list When your proposed academic supervisor is not in the list means that he or she has not been approved by the Board

[11]https://www.zotero.org/
[12]https://www.mendeley.com/
[13]https://bibdesk.sourceforge.io/
[14]https://www.overleaf.com/learn/latex/Bibliography_management_with_biblatex
[15]https://canvas.uva.nl/courses/52891/pages/submitting-your-project-proposal-in-datanose

of Examiners. You can ask the thesis coordinator to send a request for adding the academic supervisor to the list, but know that this process can take up to eight weeks, and the request may be rejected. This means you need to find different academic supervisor. If you have questions about this, contact the thesis coordinator.

My daily supervisor is not in the list Choose the option 'not in list' and add someone new.

8.5. Step 4: Embark on Your Master Project Journey

Once your proposal is approved, you are ready to embark on your master project journey in April. A good start is to set up regular (bi)weekly meetings with your supervisors. If you have an external (company) supervisor or a daily supervisor, these meetings can be more frequent, ranging from 2 to 4 hours of supervision per week.

During the three months of your project, two important events take place: the kick-off meeting and the midterm presentations. These will be explained in the next sections. Additionally, Masterproject Software Engineering organises weekly stand-ups where you can share your progress and frustration, and ask questions or tips for overcoming any roadblocks. In our experience, students who attend these weekly stand-up have a higher chance of defending their thesis in time, with a high grade.

8.5.1. Kick-off Meeting – April

Typically we kick-off the thesis journey in April with a workshop, to get you into the thesis mood. You do not need to prepare anything for the event itself, only bring your research questions. The kick-off event is usually followed by drinks.

8.5.2. Midterm Presentations – May

In May, halfway your project, you are going to present your current results, and future plans. This is an opportunity to reflect on the past 1.5 months, assess how well you are adhering to your timeline, and ensure you are moving in the right direction. This presentation will also help prepare you for your future defense, and practice your presentation skills. You will have around 10 minutes (subject to change) and should leave some time for questions. It is a good idea to invite your supervisor(s)!

8.6. Step 5: Write your Thesis

By the end of your project, you also have to write a thesis using the provided thesis template on Overleaf[16]. Though it is advisable to do this during the project, many people will leave this part to the end.

[16] https://www.overleaf.com/read/zgjhspjzfjyw#615a77

How you go about this is your own decisions. A typically way is to establish an outline with your supervisor first. Then, begin filling in the details incrementally. Just keep writing without worrying about it being perfectly written. The key is to get as much down as possible so you do not forget any details. It is easier to revise and improve a rough draft then to start from a blank page.

Keep track of your sources to ensure correct citations and avoid plagiarism, use spell checks for grammar, punctuation, and consistency. If you want help or advice to improve your academic writing skills, the 'Help with academic writing'[17] page offers a range of tips to improve on language, style, referencing, and plagiarism. Additionally, the Writing Center[18] provides free courses, workshops, and individual tutoring sessions for all UvA students. Support is available in both English and Dutch. You can also join the Guided Study Sessions[19], and be sure to check out their thesis sessions.

The UvA Scripties Database[20] has an overview of thesis written by Software Engineering students, which you can use as an example.

8.7. Step 6: Defend your Thesis

You finish your thesis project with a public defense. Typically this happens between end of block 6 (June) and the end of the academic year (August). It may be a difficult to find the time for a defense, as some people (your supervisor, your second reader) may be on holiday, so do plan accordingly. In addition, if you want to join the graduation ceremony, and not have to pay for another year of tuition fees, it is important to plan your defense before the last week of August, as processing your grade may take some time.

If your supervisor agrees the thesis is ready for defense, you can start organising your defense. Here are the steps to take before your defense:

1. Find a second reader together with your supervisor (and/or ask for the thesis coordinator help).
2. Confirm with the thesis coordinator that the defense committee fits the SE master expectations.
3. The committee (supervisor, second reader and, where applicable, daily supervisor) and you agree on a date and time for the defence.
4. Send a request to the programme coordinator via `master-se-science@uva.nl` to obtain a room for that date. Alternatively, for an online defense, you can request a Zoom room when emailing the defence details as described in the next step. Family and friends are welcome!
5. Send the thesis coordinator (cc your academic supervisor) an email with the details of your defence:
    - Names of supervisor(s) and second reader, and their affiliation,

[17] https://student.uva.nl/en/topics/help-with-academic-writing
[18] https://intt.uva.nl/language-training/writing-centre/writing-centre.html
[19] https://student.uva.nl/en/topics/guided-study-sessions
[20] https://scripties.uba.uva.nl/search?opleiding=software+engineering

- Location (or ‘online’ if you need a Zoom room),
- Date of sending the thesis to the committee,
- Date and time of the defence,
- Title of the thesis.

6. Send your final thesis to the committee and upload it to DataNose. This is usually done one week before the defense, but your committee may require you to submit it earlier or allow you to submit it later (but not after the defense).

The final grade will be determined by the academic supervisor and the second reader during the defense, according the rubric as shown in appendix A. It is a public ~20 min presentation, followed by questions from the committee. The goal of the questioning is to assess your understanding of the project and the knowledge you have gained. While, in theory, the questioning can continue until the committee is satisfied, it usually lasts about 30 minutes. After the questioning, the committee will withdraw to deliberate to determine the final grade. They will return to the room to announce their assessment, either in public or private if you choose to do so.

8.8. Step 7: Graduation and Getting your Diploma

Congratulations, you have reached the final step! After your grade has been established in the system (and assuming this was your last course) *you have to actively apply for your diploma*. Do not forget this, as you cannot apply for your degree when you are unenrolled from the programme. If this happens, you will have to wait a full year before you can apply for your degree. When in doubt, reach out to the Education Desk via `servicedesk-esc-science@uva.nl` or the programme coordinator via `master-se-science@uva.nl`.

For more information on graduation procedure and dates, please have a look at the student guide on ‘Applying for your degree certificate’[21].

When you apply for your diploma you will also be asked if you want to participate in the ceremony in October. If you do not want to do this, you can pick up your degree at a different moment, or have it send to you by post.

Additionally, do not forget to register for our Alumni mailing list[22], and join our Linkedin community[23]! It is a great way to stay connected with your fellow students and teachers. Keep an eye out for opportunities and events for alumni that we share there!

[21] https://student.uva.nl/en/topics/applying-for-your-degree-certificate
[22] https://list.uva.nl/postorius/lists/msc-se-alumni.list.uva.nl/
[23] https://www.linkedin.com/groups/9503366/

A. Master Project SE Rubric

Assessment of the research, the thesis and the presentation

Research (**daily supervisor**)	excellent	good	satisfactory	sufficient	insufficient	n.a.	weight
Theoretical knowledge							
Technical skills							
Independence / initiative							
Originality of the contribution							
Work attitude and accuracy							
Embedding in existing research							
Cooperation							

Grade:

Remarks:

Thesis (Supervisor + Commission)

Abstract							
Context							
Scientific question							
Contents							
Use of literature							
Structure							
Lay-out							

Grade:

Remarks:

Oral Presentation and Defence (Supervisor + Commission)

Context							
Contents							
Media use							
Narrative style							
Discussion / defence							

Grade:

Remarks:

Explanation of the assessment criteria

A final assessment discussion will take place between student and supervisor, in which the strong and weak points of the student's performance are discussed and the overall grades are motivated by the supervisor. The assessment criteria may be used as a guideline of what aspects of research and thesis work are generally considered as important in arriving at a final grade

Clarification of the terms

Research Work

Theoretical knowledge: did the student acquire the knowledge needed to carry out the project?
Technical skills: did the student show good experimental, programming and/or mathematical skills?
Independence/ initiative: did the student take initiatives of his/her own to carry out the project, and could he/she make progress in the (temporary) absence of close supervision?
Original contribution: did the student make an original contribution to the project?
Working attitude and accuracy: how was the overall working attitude of the student? Did the student work meticulously?
Embedding in existing research: did the student embed his work well in existing research; did he consult the relevant literature?
Cooperation: did the student actively participate in work discussions? How was the cooperation with other group members during the research? How were the student's communicative skills?

Thesis

Abstract: does the abstract contain all elements (scientific question and main conclusions) and is it written in a clear way?
Context: was the subject placed in a correct scientific context, with proper referencing of the prior work? If applicable, was the relevance for society well recognised (technological aspects, ethical aspects, historic context, or environmental aspects). Is the description of the context readable for a non-expert in the field?
Contents: does the thesis give an accurate and precise description of the subject? In case of a master's thesis: has the contribution of the student been indicated explicitly?
Scientific question: did the student properly describe the scientific question and was this question answered in a clear way?
Use of literature: is the quality and quantity of the literature sufficient? Is the literature cited adequately in an accurate list of references?
Structure: is the thesis clearly written and structured? Do the abstract and the concluding section contain the important results obtained, and is there a discussion of possible future work?
Lay-out is there a proper use of figures and graphs? Was the overall layout appealing? Is the use of the

Oral presentation and defence

Context was the research placed in a correct scientific context, with proper referencing of the prior work? Is the description of the context understandable for a non-expert in the field?
Contents: does the presentation give an accurate and precise description of the work? Has the contribution of the student been indicated explicitly? Was the scientific question presented clearly?
Media use: did the student correctly use slides, PowerPoint, animations, etc.?
Narrative style: how was the narrative style of the student, including the nonverbal communication?
Discussion / defence: were the questions answered correctly?

Grading

Each of the three components will be graded separately. There is no hard and fast rule for the relative weight of components. Suggested approximate weights are: Research 50%, thesis: 35 %, presentation and defence: 15%

References

[1] S. Easterbrook, J. Singer, M.-A. Storey and D. Damian. 'Selecting Empirical Methods for Software Engineering Research'. In: *Guide to Advanced Empirical Software Engineering*. Ed. by F. Shull, J. Singer and D. I. K. Sjøberg. London: Springer London, 2008, pp. 285–311. ISBN: 978-1-84800-044-5. DOI: 10.1007/978-1-84800-044-5_11. URL: https://doi.org/10.1007/978-1-84800-044-5_11.

[2] H. Washizak, ed. *Guide to the Software Engineering Body of Knowledge (SWEBOK Guide), Version 4.0*. IEEE Computer Society, 2024. URL: www.swebok.org.

Acronyms

AI	Artificial Intelligence
API	Application Programming Interface
BDD	Behaviour-driven development
CI/CD	Continuous Integration/Continuous Delivery
DSL	Domain-Specific Language
LLM	Large Language Model
MVC	Model-view-controller
PMP	Preparation Master Project
REST	Representational State Transfer
RUP	Rational Unified Process
SMASH	Statistics and Mathematics Advice and Support Hub
SUT	System under test
SysML	Systems Modelling Language
UML	Unified Modelling Language